

SCAPPOOSE, OR, USA

WMO: 726836

Lat: 45.773N

Lon: 122.861W

Elev: 55

StdP: 14.67

Time Zone: -8.00 (NAP)

Period: 01-19

WBAN: 04201

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.6	26.1	13.7	11.0	28.7	18.8	14.2	30.6	16.5	47.4	13.5	47.8	2.5	260	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.9	92.2	67.6	88.0	66.5	83.9	65.2	69.5	87.2	67.8	84.8	66.1	81.7	8.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	86.5	75.6	61.3	81.3	73.1	59.9	77.1	71.8	33.5	87.4	32.0	84.5	30.7	82.0	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.4	10.9	DB	16.8	99.8	4.6	3.3	13.5	102.2	10.8	104.1	8.2	106.0	4.8	108.4	(4)
(5)			WB	15.7	73.6	4.4	2.1	12.5	75.0	9.9	76.3	7.5	77.4	4.3	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	40.0	42.2	46.1	50.4	57.0	62.0	68.1	68.4	62.6	53.2	44.8	39.3		
	DBStd	11.69	6.16	5.15	4.98	5.39	6.08	6.13	5.43	5.32	5.81	5.20	6.01	6.22		
	HDD50	1280	313	222	140	58	10	0	0	0	1	27	175	334		
	HDD65	4758	776	639	584	439	257	131	27	23	113	367	605	797		
	CDD50	2338	2	3	21	69	228	359	560	570	378	126	20	2		
	CDD65	343	0	0	0	1	10	40	122	128	41	1	0	0		
	CDH74	4640	0	0	1	52	247	585	1530	1563	632	30	0	0		
	CDH80	1833	0	0	0	9	70	223	640	665	223	3	0	0		
(14) Wind	WSAvg	4.1	2.9	3.2	3.9	4.2	4.5	5.0	6.0	5.3	4.2	3.1	3.2	3.1		
(15) Precipitation	PrecAvg	51.7	8.2	5.1	5.2	4.2	3.0	2.0	0.5	0.9	1.5	4.4	8.2	8.9		
	PrecMax	76.7	16.5	13.1	9.1	8.8	5.9	4.1	2.3	3.5	3.9	8.1	17.9	17.7		
	PrecMin	39.8	2.9	0.0	1.4	1.2	0.0	0.5	0.0	0.0	0.0	1.0	2.8	4.7		
	PrecStd	9.5	3.1	2.9	1.9	2.1	1.6	0.9	0.5	0.8	1.1	1.9	3.8	3.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	61.3	70.8	80.6	86.5	93.5	98.4	97.5	92.1	78.7	63.5	57.1		
		MCWB	52.8	49.9	55.0	60.8	62.9	69.9	69.7	67.7	66.1	59.7	56.9	54.0		
	2%	DB	54.1	57.0	64.0	72.8	81.1	86.4	91.5	92.1	86.2	71.6	59.6	53.3		
		MCWB	50.4	48.0	52.2	56.9	62.3	66.0	67.5	67.6	63.8	58.5	55.0	50.4		
	5%	DB	51.5	54.0	59.3	66.0	75.0	80.8	87.0	87.6	81.4	66.7	56.5	50.3		
		MCWB	48.6	47.7	50.2	53.3	59.9	63.8	67.0	67.0	63.3	57.0	52.2	47.6		
	10%	DB	48.8	51.1	55.5	61.4	70.0	75.3	82.3	82.5	76.4	62.9	54.2	47.7		
		MCWB	46.0	46.6	49.0	51.6	57.6	61.4	65.3	65.2	61.7	55.9	50.9	45.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	53.1	57.4	62.3	66.5	70.9	73.0	71.5	69.3	64.0	59.2	54.7		
		MCDB	56.5	56.2	67.7	78.6	82.9	89.5	93.7	91.1	85.3	71.5	61.5	56.8		
	2%	WB	51.2	51.0	54.0	57.8	63.4	67.4	69.5	69.3	66.2	60.7	55.6	51.1		
		MCDB	52.7	54.7	59.9	69.8	78.0	84.2	86.6	87.3	81.1	67.4	58.4	52.6		
	5%	WB	48.5	48.7	51.9	55.0	61.0	64.5	67.7	67.6	64.2	58.6	53.4	48.0		
		MCDB	50.5	52.2	57.4	63.6	72.8	78.8	84.3	84.8	77.7	64.2	55.5	49.8		
	10%	WB	46.4	46.8	50.1	52.7	58.7	62.2	65.8	65.8	62.4	56.8	51.3	45.9		
		MCDB	48.2	50.1	55.2	59.5	68.7	73.9	81.4	81.3	73.9	61.6	53.6	47.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	15.3	17.3	19.2	20.6	21.1	25.0	25.9	24.2	18.7	14.3	10.8		
		MCDBR	14.6	21.2	24.8	30.1	31.8	31.2	33.0	33.8	33.4	26.9	16.9	12.9		
		MCWBR	10.9	13.5	14.5	15.9	16.1	14.7	14.0	13.8	14.8	14.9	11.6	9.8		
	5% WB	MCDBR	12.2	15.7	19.9	25.6	27.5	28.7	30.1	31.1	28.7	20.4	12.8	11.0		
		MCWBR	10.0	10.7	12.7	14.6	14.7	14.2	13.9	13.8	13.8	12.6	9.8	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.316	0.333	0.361	0.377	0.365	0.350	0.357	0.343	0.332	0.325	0.327		
	taud		2.525	2.551	2.514	2.398	2.372	2.421	2.488	2.479	2.507	2.548	2.525	2.463		
	Ebn at Noon		250	277	286	285	282	286	289	283	278	266	246	232		
	Edh at Noon		21	24	29	35	37	36	33	32	29	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		361	627	913	1332	1695	1871	2123	1814	1355	783	422	296		
	RadStd		59	88	121	125	161	178	151	125	120	97	57	45		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+203	N/A	
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	+50	

Nomenclature: See separate page